Comparing SHAP Explanations and Symbolic Regression for Interpretable Machine Learning

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Abstract

Two approaches to making machine learning models more interpretable are post-hoc explainability methods and symbolic regression. While post-hoc methods take an already created model and explain its behaviour, symbolic regression produces an equation that itself is the model, making the explanation intrinsic. This project aimed to compare SHapley Additive exPlanations (SHAP) of Random Forest, XGBoost and MLP models with equations produced by PySR.

The datasets were a dimensionless Feynman I.9.18 benchmark dataset as well as a body-fat dataset of 252 observations. The comparison focused on correctness, completeness, continuity and compactness, with predictive performance providing context. These were examined using feature replacement tests, input perturbations, SHAP attribution coverage and symbolic expression structure.

For the Feynman dataset, PySR recovered the generating equation and had the lowest observed prediction error. On body fat, PySR had lower observed prediction error than Random Forest and XGBoost, but higher error than MLP. Correlations between the SHAP rankings and rankings derived from feature replacement impact had positive medians. However, retaining inputs covering 90% of absolute SHAP attribution usually did not preserve predictions within the chosen tolerance when the remaining inputs were replaced. SHAP values showed high directional similarity on average under small input perturbations that produced small changes in predictions.

The comparison showed differences in what the two approaches explain. SHAP explained feature contributions to individual predictions, while PySR provided an equation that, together with input preparation, represented the complete prediction rule. Understanding this distinction and its limits can help inform the choice of explanation method.

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1 Introduction

1.1 Motivation

Machine learning has become widely adopted across many fields in both industry and scientific research. While these models can be highly accurate, they are also often internally opaque [1]. The opacity of the models' internals may seem a reasonable trade-off for their high accuracy. However, when they are to be used in situations where the cost of error is high, how can we trust something we cannot verify, challenge or interrogate? High predictive accuracy may not be enough for us to trust a model in situations of serious consequence, such as in healthcare or judicial systems [2]. Explainable artificial intelligence (XAI) is a field which aims to address this problem. It seeks to provide details or reasons that make a model's functioning clear or easy to understand for an audience, with the goal of improving trust in the model's outputs [1].

Two approaches to XAI are making models which are intrinsically interpretable and making black-box models which we try to explain afterwards using some explanatory method (post-hoc) [1]. The dominant approach has been the latter, using a secondary model to explain an already trained black-box model [1, 3]. SHAP (SHapley Additive exPlanations) falls into this category [4]. PySR is a symbolic regression (SR) algorithm which follows the first approach, producing a mathematical equation that is itself the predictive model [5]. Despite their similar goals, the literature reviewed for this project contains few systematic comparisons of the explanation quality offered by post-hoc methods and symbolic regression across multiple fields. This comparison can help people in both research and industry roles understand the advantages and limitations of the two approaches.

1.2 Aim

This study aims to compare SHAP explanations with equations produced by PySR to identify the strengths and limitations of each approach and inform the choice between post-hoc explanation and symbolic regression. SHAP explanations were generated for Random Forest, XGBoost and MLP models across two datasets, a dimensionless Feynman I.9.18 problem where the generating equation is known, and an observational body-fat dataset. These explanations will be compared with equations generated by PySR on the same datasets. The explanations will be assessed on the properties of correctness, completeness, continuity and compactness as they are described in Nauta et al. Predictive performance will also be supplementary to these properties. Notably, the aim is not to find the best model, but to compare the explainability approaches of SHAP and SR.

From this comparison the following questions will aim to be answered:

1. What do evaluations of correctness, completeness, continuity and compactness reveal about the SHAP explanations and PySR equations?

2. How does the evidence differ between the known-equation Feynman benchmark and the observational body-fat dataset?

3. What advantages and disadvantages of SHAP explanations and PySR equations emerge from their evaluation on the two datasets?

The findings aim to help researchers and practitioners make informed choices between the two approaches within the contexts examined.

1.3 Contributions

This study makes two main contributions. The first is an empirical comparison of local SHAP explanations with equations produced by PySR on the dimensionless Feynman I.9.18 problem and the observational body-fat dataset. This comparison provides evidence of the strengths and limitations of the two approaches within the contexts examined. It considers both the information SHAP provides about individual fitted-model predictions and the global prediction rule provided by a PySR equation.

The second contribution is the practical application of correctness, completeness, continuity and compactness to the two explanation forms. These properties were assessed through feature replacements, small input perturbations, attribution coverage and checks of the equations and their structure. Since SHAP explanations and PySR equations describe different aspects of their respective models, the evaluations were chosen to suit each explanation form. This allows the approaches to be considered through the same properties, while recognising that the measurements do not always allow a direct numerical comparison.

The findings support these contributions. PySR recovered the known generating equation for the Feynman problem, showing equation recovery in the controlled setting examined. In the preservation tests, SHAP-ranked feature subsets preserved fitted-model predictions better on average than matched random subsets. However, retaining features which covered 90% of absolute SHAP attribution did not ensure that predictions were reconstructed within the chosen tolerance. This shows that a compact summary of feature contributions does not necessarily provide enough information to closely reproduce a prediction.

2 Background and Related Work

2.1 Explanation forms and scope

Explainable artificial intelligence concerns the information needed to make a model's functioning clear or easy to understand for an audience [1]. However, explanations can provide this information in different forms, and the form determines what can be learnt from them. In this study, a SHAP explanation is a vector of feature contributions for one fitted-model prediction. Rankings and plots aggregated across these vectors summarise the local explanations, but do not become a single rule which can be used to make predictions. A PySR equation provides a different form of explanation: it is a global prediction rule which can be evaluated at any input within its domain. Together with the necessary input preparation and output mapping, the equation describes how the symbolic model produces its predictions. The distinction considered here is therefore between local feature attributions and an explicit prediction rule, rather than a fixed distinction between the terms interpretability and explainability. Understanding these forms provides the basis for considering how their quality can be evaluated.

2.2 SHAP and the predictive models

Rather than constraining model architecture in service of transparency, post-hoc explainability methods are applied to an existing black-box model to explain its behaviour [2]. This separates the explanation procedure from the construction of the predictor, allowing information about an already trained model to be obtained without replacing it. It does not, however, mean that every post-hoc method fits a separate explanatory model. Local Interpretable Model-agnostic Explanations (LIME) is one example which approximates a predictor locally with an interpretable surrogate [6]. SHAP provides another approach, assigning contributions to the features involved in an individual prediction [4]. While LIME is relevant to the development of post-hoc explanation, SHAP was the only post-hoc method evaluated in the completed study.

The SHAP explanations in this study concern three fitted regression models: Random Forest, XGBoost and multilayer perceptron (MLP). Random Forest is a tree-ensemble predictor, while XGBoost is the boosted-tree predictor used in the comparison. The MLP is a neural-network regressor, implemented here with a hidden layer whose width and activation were considered during model selection. These models provide different ways of learning the relationship between inputs and an output, but each ultimately returns a prediction for a given observation. SHAP supplies a common local feature-attribution form through which these predictions can be examined. This common form is useful for organising the comparison, though it does not make the predictors equivalent or remove differences between the explanation procedures used for them. The fitted configurations, preprocessing and model-selection procedures are described in Methodology; the background needed here is their role as the predictors being explained.

SHapley Additive exPlanations (SHAP) assigns a feature a contribution by averaging its marginal contribution across possible feature coalitions or orderings [4]. A marginal contribution is the change associated with including that feature in a coalition, relative to the coalition without it. Averaging considers the different combinations in which a feature can contribute, rather than assessing it in only one fixed ordering. To calculate these quantities, the explanation procedure must specify how features absent from a coalition are treated. For an individual prediction, the resulting SHAP value is a signed contribution relative to the explainer's baseline: a positive value contributes towards a prediction above the baseline, while a negative value contributes towards a prediction below it. In the additive representation, the baseline and the feature contributions reconstruct the explained output. This is an account of a fitted-model prediction, rather than an assessment of its agreement with the observed target. Likewise, a contribution does not establish the causal effect of changing a measurement in the real world. Its interpretation remains tied to the predictor and the explanation setup.

Lundberg and Lee place SHAP within a class of additive feature-attribution methods and identify three desirable properties: local accuracy, missingness and consistency [4]. Local accuracy requires the additive explanation to match the model output for the observation being explained. It should therefore be distinguished from predictive accuracy, which concerns agreement with the target. The formal consistency property also differs from the continuity property evaluated in this study, which concerns explanation changes under nearby inputs. These theoretical properties provide a rationale for examining SHAP, but do not settle every

question about its practical use. In their reported comparisons, Lundberg and Lee found computational and human-intuition advantages for the approaches they introduced [4]. Such findings provide context for selecting SHAP; they do not establish its superiority for every model, dataset or audience. In particular, the KernelSHAP comparisons discussed in the literature should not be treated as findings of this project, whose primary explanations used TreeSHAP and exact SHAP.

Computing exact Shapley values requires consideration of feature combinations, making the calculation expensive as the number of features increases. KernelSHAP provides a model-agnostic calculation through weighted regression, while TreeSHAP exploits tree structure to calculate Shapley values efficiently [4, 7]. In this study, the Feynman Random Forest and XGBoost models use path-dependent TreeSHAP, whereas the MLP uses exact SHAP with an empirical background. The primary body-fat explanations use empirical-background exact SHAP for all three predictors. They concern thirteen raw measurements, with two derived ratios recalculated inside the predictor before its fifteen model inputs are formed. Both representation and background matter to the interpretation: Feynman attribution differences can reflect the explanation conventions as well as the fitted models, while the body-fat comparison shares its primary representation and empirical background. Exact enumeration removes coalition-sampling approximation for the chosen setup. It does not remove dependence on the background or make the contributions causal.

Attribution methods also face practical limitations which require evaluation beyond their formal properties. Salih et al. discuss the effects of the underlying model and feature collinearity on SHAP and LIME, showing why their outputs require care in interpretation [8]. An attribution which appears reasonable is not necessarily a faithful account of the model's behaviour [9]. One way to examine this relationship is to compare attributed importance with changes in the fitted prediction after feature replacement. However, replacing one feature while keeping others fixed can create unrealistic combinations when measurements are dependent. These tests consequently provide evidence under a specified replacement procedure, rather than unrestricted evidence about the world. The need for such checks does not mean that post-hoc explanations must be unfaithful by definition. It means that what an attribution describes, and the evidence used to assess it, must be made explicit.

2.3 Symbolic regression and PySR

Symbolic regression (SR) is a supervised learning task in which the space of mathematical expressions is searched to find an equation relating inputs to an output [5]. Unlike an explanation added to a fitted black-box model, the expression produced by SR is itself the predictor. It can be evaluated directly to obtain a prediction, and the operations through which the inputs affect that prediction are available for inspection. Genetic-programming approaches represent expressions as computational trees and evolve candidate expressions, providing a way to search over their structure as well as their parameters [10]. The relevance of SR to this study is therefore not simply that it produces another feature-importance measure. It provides a global mathematical rule describing the fitted relationship. This can make interactions accessible through the expression's functional form, though having access to those operations does not necessarily make a complex expression easy to understand.

PySR is the genetic-programming-based SR implementation used in this study. It treats

equation discovery as a search over candidate mathematical expressions, combining variables, constants and permitted operators [5]. The available operators and restrictions on expression size constrain which relationships can be represented, while parsimony settings make expression complexity relevant to the search. Predictive fit and complexity consequently provide different information when inspecting candidates: an expression may fit the data closely but be difficult to follow, while a shorter expression may be easier to inspect but omit relationships needed for accurate prediction. Complexity is relevant to interpretation, but a small expression is not by itself evidence that a person understands it [9]. For the comparison undertaken here, PySR's value is that its mathematical operations and parameters can be examined as one executable prediction rule. The operator choices, search budgets and selection procedures used to obtain the reported equations are methodological choices, and are described in the following chapter.

Despite this explicit representation, SR faces limitations in both search and validation. Genetic-programming methods can struggle with larger expression spaces and higher-dimensional datasets, and their behaviour depends on the restrictions placed on the search [10]. Domain knowledge can help constrain candidate relationships, but those constraints also influence what can be discovered. A further difficulty concerns identifiability: whether the underlying relationship can be uniquely recovered from the available observations [11]. Where a generating equation is known, a discovered expression can be compared with that reference. Where it is unknown, good predictive fit alone does not establish that the expression represents the real process rather than a relationship fitted to the available data. This distinction is especially relevant to the observational body-fat case. SR exposes the fitted rule, but the correctness of that rule as an account of the world, and any causal interpretation of it, still require separate evidence.

2.4 Evaluating explanation quality

Evaluating an explanation is complicated by the fact that explanation quality depends on what is being explained and who needs to understand it. The relevant measures can differ with the application, explanation form, data and background knowledge of the audience [9]. Predictive performance does not resolve this problem. A model can predict accurately while an explanation gives a poor account of how it produces those predictions; conversely, an explanation may faithfully describe a model whose predictions are inaccurate. Likewise, a plausible explanation may reflect what a user expects rather than what the model actually does [9]. For a comparison between SHAP and native equations, these distinctions make it necessary to define the properties being assessed and the evidence appropriate to each form. Explanation quality is therefore considered property by property, rather than reduced to a single score.

Nauta et al. discuss the taxonomy of Doshi-Velez and Kim, distinguishing application-grounded, human-grounded and functionally grounded evaluation [9]. Application-grounded evaluation uses domain experts on real tasks, human-grounded evaluation uses people on simplified tasks, and functionally grounded evaluation uses computational measures without requiring participants. Each provides a different kind of evidence. Evaluations involving people can examine how explanations are understood or used, but perceptions of plausibility need not establish faithfulness. Computational measures permit systematic checks, but their

relevance depends on how well they represent the intended property. This study is primarily functionally grounded, combining computational diagnostics with audits of the native symbolic rules. The body-fat survey provides supplementary feedback from two respondents on the presented Random Forest and XGBoost SHAP summaries and the PySR equation; MLP was not included. These responses concern reported understanding and preference, rather than demonstrated performance on an interpretation task or representative evidence about users more broadly.

To organise explanation evaluation, Nauta et al. propose twelve quality properties, referred to as the Co-12, across content, presentation and user dimensions [9]. This framework treats explanation quality as multidimensional rather than as a binary property which an approach either possesses or lacks. The completed project examines four properties: correctness, completeness, continuity and compactness. These concern alignment with the predictor, coverage of its outputs or reasoning, behaviour under nearby inputs, and the amount of information presented. They provide a common set of questions for the comparison, but the explanation forms still require different operational measures. A test suitable for a local attribution vector cannot automatically be interpreted as a test of a global equation. Assessing these four properties also does not amount to assessing all twelve, or establishing overall human interpretability, trust or usefulness. The following discussion sets out what each property means for the evidence considered here.

Correctness concerns how faithfully an explanation describes the predictive model it is intended to explain [9]. For SHAP, this can be examined by comparing a local absolute-attribution ranking with the ranking of prediction changes after individual features are replaced. Agreement provides evidence about fitted-model faithfulness under the chosen replacements. Disagreement needs care: SHAP averages contributions across coalitions, whereas single-feature replacement changes one input while retaining the other observed values. Interactions and the replacement convention can therefore affect the comparison. For PySR, a different check is needed: whether the displayed equation, together with its preparation and output mapping, reproduces the fitted predictor. This is an execution audit of the disclosed rule, rather than the same empirical test as an attribution-ranking comparison. The Feynman case also permits a separate assessment of whether that rule agrees with the known generator. Faithfulness to a fitted model, correct execution of its disclosed rule and agreement with a generating relationship are distinct questions. The body-fat data provide no equivalent known equation against which to establish recovery, and an explicit rule should not simply be awarded a universally perfect correctness score.

Completeness requires a further distinction between output-completeness and reasoning-completeness. Nauta et al. describe output-completeness in terms of how much of the predictor's output an explanation covers, while reasoning-completeness concerns how fully it describes the internal operations and parameters [9]. For SHAP, output-completeness can be investigated by retaining the highest-ranked observed features and replacing the remainder to assess prediction reconstruction. Replacing the highest-ranked features instead assesses how their removal disrupts the prediction. Matched random orders provide a reference for whether the attribution ordering is informative. These tests differ from counting the features which cover a chosen proportion of absolute attribution: covering 90% of attribution magnitude does not ensure reconstruction within a specified tolerance. Similarly, additivity of the full SHAP vector accounts for the explained output, but does not disclose every oper-

ation inside the predictor. A native PySR equation, together with its required preparation and output mapping, can instead expose the complete fitted prediction rule. This concerns reasoning-completeness with respect to that model. Measured reconstruction or disruption and disclosure of a mathematical rule consequently answer different questions; they are not equivalent numerical scores. Neither establishes that the explanation is a complete account of the real-world phenomenon, particularly human body composition.

Continuity concerns whether small input changes produce appropriately small explanation changes when the fitted response remains nearly the same [9]. For SHAP, this can be examined by perturbing an input and comparing the original attribution vector with the perturbed one after checking response similarity. Direction and magnitude both matter. Cosine similarity describes agreement in direction, while an attribution-distance measure describes changes in the contribution values; high cosine does not necessarily mean unchanged attributions. This nearby-input question differs from repeatability for identical inputs, which Nauta et al. discuss under consistency, and from SHAP's formal consistency property [9, 4]. A PySR equation presents another distinction. Its structure remains fixed across inputs, but its output can still be numerically sensitive, and some expressions have inputs at which they are undefined. Output sensitivity and the equation's domain therefore require separate consideration. Unchanged equation text does not supply a SHAP-vector continuity score or establish superior continuity. The SHAP evidence is also conditional on the examined inputs, background and response-stability filter.

Compactness concerns the size of an explanation and the need to avoid presenting more information than a reader can reasonably follow [9]. For SHAP, one measure is the smallest ranked feature subset covering a specified share of total absolute attribution. The count depends on the chosen share, so it describes compactness under a reporting convention rather than an intrinsic explanation size. For a native equation, distinct variables, operator occurrences and expression-tree nodes describe different aspects of its structure. These measures are relevant to how much information is exposed, but use different units from a local attribution feature count. A smaller node count cannot therefore be ranked directly against a SHAP feature count as proof of better interpretability. Expression size is also an incomplete proxy for understanding [12]. Finally, the size of the fitted predictor, such as tree totals or MLP parameter counts, should remain separate from the size of its displayed explanation.

The availability of a known generating relationship changes what these evaluations can establish. In controlled synthetic data, explanation evidence can be compared with a specified reference rather than inferred entirely through proxy measures [9]. The Feynman case in this study is the single dimensionless I.9.18 problem, whose known equation allows recovery to be checked. It is not an evaluation of the entire Feynman collection. The body-fat dataset provides observations from 252 men, but no known generating equation supplies an equivalent explanatory reference. Its evidence therefore concerns fitted-model behaviour, the disclosed prediction rule and representation-specific properties. Accurate prediction or an accessible equation does not establish a causal mechanism for body composition. The cases provide complementary settings with different reference evidence, rather than a design which isolates general effects of dataset characteristics. Conclusions from the controlled problem cannot simply be extended to observational data or to whole scientific fields.

2.5 Related work and scope of this study

Existing SR evaluation work illustrates both the value and limitations of combining different forms of evidence. SRBench++ extends benchmarking through synthetic tasks with known reference functions and a real-world COVID-19 modelling track involving domain-expert assessment [12]. Predictive performance, expression size and expert assessment address different aspects of the models, but the authors recognise that expression size is an incomplete interpretability proxy and expert judgments do not provide a readily generalisable evaluation. Aldeia and de França also examine explanatory methods applied to regression models, including symbolic-regression models, using the Feynman data set [13]. This is relevant because it places SHAP and symbolic predictors within an empirical evaluation setting. However, applying an explainer to a symbolic predictor is a different question from evaluating its native equation alongside SHAP explanations of other predictors. The latter concerns the information exposed by two explanation forms, rather than the performance of a common explainer when applied to different models.

Direct comparisons between interpretable models and post-hoc explanations also exist. Mahya and Fürnkranz compare approaches across six classification datasets, pairing methods which produce the same output form: GLOCALX with JRIP for rules, and MAPLE with GA 2MS for feature-based explanations [14]. Accuracy, fidelity and complexity provide comparison criteria within that design. Matching output forms helps control what is compared, but does not make the resulting measures directly transferable to local attribution vectors and native equations. Sepioło and Ligęza compare LIME and SHAP with symbolic models generated through grammatical evolution on a quadratic classification task and the Iris dataset [15]. This provides a direct example of a symbolic and post-hoc comparison, while differing from this study in its classification tasks, symbolic method and use of domain-informed representations. Existing work therefore rules out an absolute claim that no comparison has been conducted. Its relevance lies in the designs and evaluation choices it provides, rather than in establishing a universal advantage for either approach.

Taken together, these studies provide useful evaluation concepts and examples, while leaving the choice of comparable evidence dependent on the explanation forms and questions involved. This study examines local SHAP explanations of Random Forest, XGBoost and MLP alongside native PySR equations on the dimensionless Feynman I.9.18 problem and the observational body-fat dataset. Correctness, completeness, continuity and compactness organise the assessment, with predictive performance providing context. The comparison preserves the distinctions between fitted-model faithfulness, output reconstruction, disclosure of a prediction rule, stability and representation size. Its contribution is an applied examination of these properties in the two cases, rather than a first-ever comparison or a universal explanation-quality ranking. The following chapter describes the datasets, preparation, model selection and operational tests needed to interpret that evidence.

3 Methodology

3.1 Study design

The project compared SHAP explanations of Random Forest, XGBoost and multilayer perceptron (MLP) regressors with native PySR equations on two regression tasks. Correctness, completeness, continuity and compactness were assessed using the distinctions described by Nauta et al. [9]. Predictive performance provided context, and an informal body-fat survey supplied exploratory feedback. The comparison concerned fitted model–explanation combinations: both the predictor and explanation form differed. Measurements were reported separately, without a combined explanation-quality score.

3.2 Datasets and preparation

The Feynman case used the dimensionless I.9.18 gravitational-force problem. A sample of 10,000 rows was drawn without replacement from the supplied million-row file using seed 42. The physical relationship in the benchmark metadata was $F = Gm_1m_2/[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]$. The local dimensionless metadata specified the force scale Gm_1^2/x_1^2. Dividing by this scale gave the target $t = Fx_1^2/(Gm_1^2)$ and six ordered inputs: m_2/m_1, x_2/x_1, y_1/x_1, y_2/x_1, z_1/x_1 and z_2/x_1. The reference equation was:

$$t = \frac{m_2/m_1}{(x_2/x_1 - 1)^2 + (y_2/x_1 - y_1/x_1)^2 + (z_2/x_1 - z_1/x_1)^2}.$$

The analysis concerned the positive sampled domain; the equation is undefined at zero separation.

The body-fat CSV was obtained from the Kaggle Body Fat Prediction Dataset. It contained 252 men. StatLib documentation describes body density measured through underwater weighing and body-fat percentage calculated using Siri's equation, $495/\text{Density} - 450$. The project used the recorded body-fat target rather than recalculating it. Density was excluded from the predictors because it determines the target. The remaining 13 measurements were age in years, weight in pounds, height in inches, and neck, chest, abdomen, hip, thigh, knee, ankle, biceps, forearm and wrist circumferences in centimetres. No known generating equation was available for these retained predictors.

Column names were standardised and expected numeric columns were coerced to numeric values. Abdomen/chest and hip/thigh ratios were added before imputation, with zero denominators and non-finite ratios treated as missing. Rows with missing targets were dropped. Each predictor's median was calculated from the outer training split and used to fill missing inputs in either split. The supplied data contained no missing measurements, so this safeguard was inactive. Measurements were retained without outlier removal or unit conversion.

Body-fat models received 15 inputs: the 13 measurements and two ratios. Primary explanations concerned the 13 raw measurements. A prediction wrapper recalculated both ratios after masking, replacement or perturbation and passed the 15 inputs to the fitted model in saved order. Trees and PySR used unscaled inputs and direct target outputs. MLP scaling is described below.

Both datasets used an 80/20 random split with seed 42, giving 8,000 training and 2,000 historical test rows for Feynman, and 201 training and 51 test subjects for body fat. The Feynman correction introduced separate evaluation samples. Table 1 identifies the samples used by the final analyses.

Table 1: Evaluation samples

Analysis	**Feynman**	**Body fat**
Final predictive performance	2,000 fresh final rows	51 test subjects
Local faithfulness and completeness	50 fresh final rows	51 test subjects
Global equation alignment	2,000 historical test rows	No known-equation reference
SHAP continuity	50 historical test rows × 20 repeats	12 test subjects × 10 repeats
Prediction noise	2,000 historical test rows × 100 repeats	51 test subjects × 100 repeats
SHAP compactness	2,000 historical test rows	51 test subjects

3.3 Model training and selection

Random Forest and XGBoost used 100 randomly sampled configurations followed by a grid around the best configuration. Both stages used five-fold cross-validation within the outer training split, scored by negative mean squared error (MSE). Selected settings were fitted on the full training split with seed 42. Random Forest enabled out-of-bag scoring; XGBoost used squared-error regression. Test data supplied to the original XGBoost evaluation log did not control early stopping. The saved Feynman trees were subsequently fitted directly using the settings from the completed searches. Body-fat analysis loaded the saved HPC fits. The selected forests contained 277 trees of maximum depth 20 for Feynman and 419 of maximum depth 8 for body fat. XGBoost used 603 trees of depth 4 and 77 of depth 2, respectively.

Each MLP had one hidden layer. Input and target standardisation were fitted within each cross-validation training fold, and predictions were returned in original target units. A 60-configuration random search was followed by grid refinement, using five-fold negative-MSE scoring. Candidate widths were 16, 32 and 64 for Feynman, and 8, 16 and 32 for body fat; activations were ReLU and tanh. Both searches varied regularisation. Feynman used Adam and varied the initial learning rate; body fat used L-BFGS. Fits allowed 2,000 iterations, tolerance 10^{-5}, and 50,000 function evaluations for L-BFGS. Adam used a 10% internal training-validation split and stopped after 30 iterations without improvement. Final tanh networks had 64 and 8 hidden units, with regularisation parameters approximately 0.00174 and 11.82. The Feynman learning rate was approximately 0.00427. Training-only cross-validation at seeds 42, 1729 and 2026 checked sensitivity without selecting another seed.

PySR [5] was fitted directly to each target. Searches allowed addition, subtraction, multiplication, division and squaring, with maximum expression size 20. Population-size/number

pairs were (33, 15), (50, 20), (100, 40) and (200, 80); iterations were 200, 400 and 1,000; parsimony values were 0.001, 0.0032, 0.01 and 0.05; adaptive scaling values were 500, 1,000, 3,000 and 10,000. These defined 192 configurations per dataset. Searches used seed 42 and multithreading, which did not guarantee deterministic equation evolution.

Body-fat PySR divided the 201 training subjects into 150 fitting and 51 validation subjects using seed 42. The completed search stored 2,917 equations across 192 configurations. Selection used the lowest validation RMSE, with complexity as the secondary criterion. The selected equation was not refitted on all 201 training subjects. Its configuration used population size 50, 20 populations, 200 iterations, parsimony 0.01 and adaptive scaling 1,000.

Feynman searches retained the minimum-training-loss equation from each configuration fitted on 8,000 rows. The original configuration ranking used historical test MSE. The correction therefore reselected among the 183 available completed candidates using 2,000 fresh validation observations, seed 9020801. Physical variables were independently sampled uniformly: m_1, m_2, x_2, y_2 and z_2 from [1, 2], and x_1, y_1 and z_1 from [3, 4], following the benchmark metadata. Ratios and targets were then calculated. Non-finite predictions received infinite validation MSE. Candidates within 10^{-20} of minimum MSE were ordered by complexity, then configuration label. The selected configuration used population size 100, 40 populations, 1,000 iterations, parsimony 0.0032 and adaptive scaling 500.

A selection lock was saved before generating 2,000 fresh final observations with seed 9020802. Exact input duplicates against original and validation rows were checked. All predictors remained frozen. This sample supplied final predictive performance; previously inspected test results remained historical diagnostics.

3.4 Explanation construction and checks

Feynman tree explanations used path-dependent TreeSHAP [4, 7] with training path counts encoded in the saved trees. Feynman MLP used exact enumeration of 64 coalitions with 50 training-background rows sampled without replacement, seed 42. Body-fat explanations used exact enumeration of 8,192 coalitions for all three predictors, sharing the same 50-row training background and raw-feature wrapper. The identity link preserved target units. MLP explanations included input scaling and inverse target scaling. The MLP explainer skipped masking calculations only for input values exactly matching the background, since near-equal values could change predictions.

Earlier body-fat tree explanations used one forward/reverse permutation cycle, or 27 evaluations. Calibration compared 1, 4, 16, 64 and 256 cycles across seeds 42, 1729 and 2026 against independent calculations and exact enumeration. Final computational arrays used exact enumeration. It removes permutation-sampling error for the fixed background, while retaining dependence on that background.

Checks verified feature order, subject alignment, saved-prediction reproduction and reconstruction by the baseline plus summed SHAP values. Clean local SHAP additivity checks used absolute and relative tolerances of 10^{-5} for Feynman trees, 10^{-10} for body-fat exact tree explanations and 10^{-9} for MLP exact explanations. Perturbed body-fat tree checks used 10^{-9} for both. The shared completeness routine used separate numerical validation settings, distinct from the reconstruction tolerances and continuity response-stability thresholds below. Independently constructed explainers checked unchanged inputs: 50 Feynman subjects for

trees and MLP, all 51 body-fat subjects for trees, and 12 for body-fat MLP. Mean absolute SHAP summarised contribution magnitude; beeswarms displayed signed contributions and input values.

3.5 Evaluation procedures

3.5.1 Predictive performance

For observed targets y_i and predictions $\hat{y}_i$, performance was measured by:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}},$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|.$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}.$$

RMSE and MAE were in dimensionless target units or body-fat percentage points; MSE used squared units. Input-noise diagnostics used 100 paired repeats on the original test splits at 2.5% and 5% of training-feature standard deviation. Accuracy was calculated against unchanged clean targets. Mean absolute prediction change and the response-stable proportion were reported separately from accuracy and explanation continuity.

3.5.2 Correctness

Feynman permitted a global reference check against the known equation. Each feature was independently permuted ten times across the 2,000 historical test rows, seed 42. Mean absolute changes in generating-equation output were averaged over permutations and compared with each model's mean absolute SHAP using Spearman correlation. This measured global reference alignment, separately from fitted-model faithfulness.

Local faithfulness compared absolute SHAP rankings with mean absolute changes in a frozen predictor after replacing one feature from training donors. Feynman used 50 fresh final rows selected without replacement, seed 9020803. Impacts averaged three independent 1,024-donor estimates, drawing with replacement from 8,000 training rows at seeds 42, 1729 and 2026. Budgets of 64, 256 and 1,024 were compared with the other two seeds' full-budget average. Body fat used all 51 test subjects and all 201 training donors once per subject/feature. Approximate budgets of 20, 50, 100, 200, 500 and 1,000 were checked against this finite-donor reference. Donor designs were paired across predictors, with body-fat ratios recalculated. Averaging local body-fat impacts also supplied its global summary.

Agreement used average-rank Spearman correlation and top-three overlap. Constant vectors had undefined correlations. Third-place ties shared remaining membership fractionally; overlap was the membership-vector dot product divided by three. Feature-order tie-breaking was retained as a sensitivity check. Medians, quartiles, interquartile ranges and defined/undefined counts summarised local agreement. Convergence checks compared impact ranks, top-three overlap and normalised absolute differences.

Native-rule audits evaluated literal PySR expressions with saved constants, preparation and output mapping against saved predictions. Body-fat reproduction used absolute tolerance 10^{-12} and zero relative tolerance. Feynman additionally checked literal versus simplified expressions and symbolic equivalence to the generator on its non-singular domain. These checks distinguished rule execution from equation recovery. Replacement diagnostics could create unrealistic hybrids and did not measure causal effects.

3.5.3 Completeness

For each observation and predictor, features were ordered by decreasing absolute SHAP, with stable feature-order ties. Let S_k contain the first k features and d be a whole donor row. Preservation retained selected observed values and replaced the remainder; deletion replaced selected values and retained the remainder:

$$P_k(x) = \mathbb{E}_d \left| f(x_{S_k}, d_{\overline{S_k}}) - f(x) \right|.$$

$$D_k(x) = \mathbb{E}_d \left| f(d_{S_k}, x_{\overline{S_k}}) - f(x) \right|.$$

The empirical averages covered $k = 0$ through p. Smaller preservation error indicated better retention of the prediction; larger deletion change indicated stronger disruption. Comparisons used 32 shuffled feature orders per observation at matched subset sizes, including the smallest prefix covering 90% of absolute SHAP. This coverage was not defined as 90% prediction recovery.

Feynman used the same 50 fresh rows as local faithfulness and 512 whole-row training-donor draws per observation, seed 9020810; random-order seed was 9020811. Body fat used all 51 test subjects and each of the 50 SHAP-background rows once per subset, with random-order seed 314160. Donors were paired across predictors, preservation, deletion and random references. MLP reused saved designs. Body-fat ratios were recalculated. Whole-row replacement preserved relationships within the replaced block but could break dependence between retained and replaced values.

Raw curves were retained in target units. Normalised curves divided by $A(x) = P_0(x) = D_p(x)$, the all-replaced change. Normalisation was undefined when $A(x) \leq \max(10^{-12}, 10^{-6} s_f)$, where s_f was the population standard deviation of training predictions for that model. Those cases retained raw results. Trapezoidal areas were calculated against k/p, with lower preservation area and higher deletion area indicating better performance under this intervention. Curves were neither clipped nor smoothed; reversals were counted. Random-reference plots used empirical 2.5th/97.5th percentiles across order repeats.

Prediction reconstruction used $\tau_f = \max(\eta, 0.05 s_f)$, where $\eta = \max[10^{-12}, 10^{-10} \max(1, \max_x |f(x)|, s_f)]$ over evaluated rows. The reported count was the smallest k with $P_k(x) \leq \tau_f$ for that and every larger prefix. First crossings were also saved because curves could reverse. Endpoint and SHAP-additivity checks validated the calculations.

PySR reasoning-completeness was assessed through disclosure of the native equation and necessary preparation and output transformations. This included dimensionless inputs for Feynman and original units, ratio preparation and saved median imputation for body fat. It concerned the fitted prediction rule, separately from SHAP output-completeness or a complete account of the underlying phenomenon.

3.5.4 Continuity

Inputs were perturbed as $x'_{ijr} = x_{ij} + \lambda s_{X_j} z_{ijr}$, with independent standard Gaussian components, seed 42, and $\lambda = 0.025$ or 0.05. Training standard deviations used population calculations. The same draws were paired across predictors and scaled for both levels. Noise affected six Feynman inputs or 13 raw body-fat measurements, followed by ratio recalculation. Inputs were not clipped to training ranges.

Feynman explanation continuity used 50 historical test rows sampled with NumPy seed 42 and the first 20 prediction-noise repeats: 1,000 candidate pairs per predictor/level. Body fat used 12 subjects and the first ten repeats: 120 pairs. This subset came from 12 evenly spaced pooled tree-prediction ranks plus the largest training-standardised raw-feature outlier, with duplicates removed. It was fixed before continuity assessment and reused for MLP.

Pairs were retained when $|f(x') - f(x)| \leq 0.05 s_y$, where s_y was the population standard deviation of observed training targets. Thresholds were approximately 0.01985 dimensionless units and 0.43374 percentage points. Retained counts and proportions were reported, with sensitivity at 2.5% and 10% of s_y. Each predictor could retain different pairs.

SHAP was recomputed using the same explanation convention and background. Signed-vector cosine similarity measured direction, and mean absolute attribution distance measured magnitude:

$$C = \frac{\varphi(x) \cdot \varphi(x')}{\|\varphi(x)\|_2 \|\varphi(x')\|_2}, \qquad L = \frac{1}{p} \sum_j |\varphi_j(x') - \varphi_j(x)|.$$

Two zero vectors received cosine 1; exactly one received 0. Distance retained the non-zero magnitude. Machine-epsilon checks handled near-zero values and cosine was bounded to $[-1, 1]$. Empirical 2.5th/97.5th percentiles of repeat means described perturbation variability, not population confidence intervals.

PySR output sensitivity was reported separately from attribution continuity. Domain audits checked finite values and denominator magnitudes across all 100 repeats, using near-zero tolerance $10^{-10} \max(1, \text{median}\,|q|)$ for denominator q. Body-fat audits also checked positivity, ratio denominators, training-range violations and equation gradients. These checks concerned the evaluated inputs; fixed equation structure did not supply a SHAP continuity score.

3.5.5 Compactness

Local SHAP size was the smallest ranked prefix covering 90% of total absolute attribution, with 80% and 95% sensitivity checks. Counts covered all historical Feynman test rows and all body-fat test subjects. Distributions, means, medians, quartiles and ranges were reported; totals at or below machine epsilon received count zero. Native-equation size used the literal saved structure: distinct variables, operator occurrences, signed numeric leaves and expression-tree nodes, separately from recorded PySR complexity. Fitted tree or network size and historical SHAP-on-PySR counts were kept separate. These different units were not ranked as interchangeable sizes.

3.6 Supplementary comparisons and survey

3.6.1 Native feature importance

Random Forest mean decrease in impurity and XGBoost gain were compared descriptively with mean absolute SHAP using normalised shares, rank differences, Spearman correlation and tie-aware top-three overlap. Feynman used six inputs; body fat required separate direct path-dependent TreeSHAP over 15 prepared inputs. MLP had no split-based importance comparison.

3.6.2 Exploratory survey

An informal Google Forms body-fat survey recorded two responses, providing exploratory feedback on the presented explanations. Fixed presentation order was A: Random Forest beeswarm, B: XGBoost beeswarm, C: PySR equation. MLP and Feynman were not included. Introductory instructions described the data, units and explanation reading. The equation stimulus was:

$$\text{Predicted body fat (\%)} = \frac{\text{Abdomen}^2 - 272.58493\,\text{Wrist}}{\text{Weight} + \text{Neck} - \text{Biceps}}.$$

Weight was in pounds and displayed circumferences in centimetres. Survey beeswarms used the historical permutation-SHAP archive, rather than final exact arrays. Labels showed unadjusted p-values from separate univariable regressions with intercepts on all 252 subjects, computed using SciPy. The reference table retained coefficients, correlations, R^2, t-based 95% slope intervals and Benjamini–Hochberg adjustment across 13 tests. These associations were separate from SHAP faithfulness.

Each explanation had six agreement items covering reading, contributor plausibility, directional plausibility, understanding higher/lower effects, information amount and usefulness. Ratings ran from 1, Strongly Disagree, to 5, Strongly Agree. Three comparative questions asked about ease, domain consistency and preference for use, including neutral or unable-to-assess options. Preference reasons and optional comments were recorded. Analysis displayed individual ratings and preference counts without a composite score or inferential test. Missing information remained missing. This two-response feedback had limited evidential weight: it concerned perceptions of those stimuli and did not demonstrate understanding, establish representative preferences or validate explanation quality.

3.7 Reproducibility

Python scripts, notebooks, saved predictors and audit files recorded settings, seeds, feature order, subject identities and paired designs. Training used the Pawsey workflow; final analysis consumed saved fits. Principal explanation software included SHAP 0.51.0, scikit-learn 1.8.0 and XGBoost 3.2.0. Body-fat tree training manifests recorded scikit-learn 1.9.0; saved predictions were checked after loading locally.

4 Feynman Results

The Feynman results concern the dimensionless I.9.18 problem, where the generating equation is known. This allows recovery of the relationship to be examined alongside the explanation properties. Predictive performance is reported first, followed by correctness, completeness, continuity and compactness. The evaluation samples follow those described in Section 3.2: final predictive performance uses 2,000 fresh observations, while the explanation tests use the specified fresh or historical subsets.

4.1 Predictive performance

PySR reproduced the generating relationship to numerical precision on the fresh final sample. Table 2 reports the performance of all four frozen predictors after validation selection. MLP had the lowest error among the three models explained using SHAP, followed by XGBoost and Random Forest. Its RMSE was 0.006354, compared with 0.037645 for XGBoost and 0.084252 for Random Forest. These results describe agreement with the dimensionless target and provide context for the explanation evaluations.

Table 2: Feynman predictive performance on 2,000 fresh final observations with dimensionless errors

Model	R^2	**RMSE**	**MAE**
Random Forest	0.953298	0.084252	0.055319
XGBoost	0.990676	0.037645	0.023358
MLP	0.999734	0.006354	0.001915
PySR	1.000000	2.042×10^{-16}	1.432×10^{-16}

The fresh sample was generated after the PySR selection was frozen. Selection covered the 183 completed configurations available from the planned 192-configuration search. The earlier test results were retained as historical diagnostics, since they had previously been used to rank PySR configurations. The results in Table 2 therefore provide the final predictive comparison under the corrected evaluation procedure.

4.2 Equation recovery and correctness

4.2.1 Recovery of the generating equation

The selected PySR expression simplified to:

$$\hat{t} = \frac{m_2/m_1}{(x_2/x_1 - 1)^2 + (y_1/x_1 - y_2/x_1)^2 + (z_1/x_1 - z_2/x_1)^2}.$$

Subtracting the known generating expression and simplifying gave zero. The reversed order of terms within two squared differences did not change the relationship. Literal evaluations reproduced the corresponding saved predictions with maximum differences of 4.441×10^{-16} on the historical test sample and 1.332×10^{-15} on the fresh final sample. The

simplified selected expression also passed the numerical agreement check. Together, these checks establish recovery of the reference equation on its non-singular domain.

The global SHAP comparison also showed close alignment with the known equation's permutation-impact ranking over the 2,000 historical test rows. Spearman correlation was 0.943 for both Random Forest and XGBoost, and 1.000 for MLP. All three placed the mass ratio first. The two tree models reversed the reference ordering of the next two features, while MLP matched the full ranking. This comparison concerns aggregate alignment with the generating relationship; local faithfulness to each fitted predictor was assessed separately.

4.2.2 Local fitted model faithfulness

Across the 50 fresh final observations, larger absolute SHAP values generally identified features whose replacement produced larger changes in the fitted prediction. Table 3 compares the local SHAP rankings with the donor-replacement impact rankings. Median Spearman correlation was 0.6571 for Random Forest and 0.7714 for both XGBoost and MLP. The interquartile ranges show that agreement varied between observations, with Random Forest having the widest middle range. Correlations were defined for all 50 observations for each model.

Table 3: Feynman local ranking agreement on 50 fresh final observations

Model	**Median ρ**	**Q1**	**Q3**	**Median top three overlap**
Random Forest	0.6571	0.3714	0.8286	0.6667
XGBoost	0.7714	0.5429	0.8286	0.6667
MLP	0.7714	0.5571	0.8286	0.6667

The median top-three overlap was two features out of three for every model. Local agreement was not uniformly positive: correlations were negative for 3/50 Random Forest observations and 4/50 XGBoost observations, with minima of -0.4286 and -0.2571, respectively. MLP had no negative cases among its 50 observations and a minimum of 0.0857. These values measure ranking agreement rather than the proportion of an explanation that is correct. Individual ratio replacements can break relationships between inputs that share a denominator. Disagreement under this replacement procedure does not by itself show that SHAP was calculated incorrectly. The tree explanations also used path-dependent TreeSHAP, whereas MLP used an empirical background, so differences between them can reflect the explanation convention as well as the predictor.

4.2.3 Supplementary native feature importance

The native tree-importance rankings broadly agreed with mean absolute SHAP. Spearman correlation was 0.943 for Random Forest and 0.829 for XGBoost, with complete top-three overlap for both. These comparisons describe agreement between two summaries of the fitted trees. They provide supplementary context rather than an independent reference for explanation correctness.

4.3 Completeness

SHAP-ranked feature subsets preserved fitted-model predictions better on average than equally sized random subsets. At the smallest local subset covering 90% of absolute attribution, mean normalised preservation error was 0.3027 for Random Forest, 0.3665 for XGBoost and 0.3104 for MLP. Each was lower than its matched random reference in Table 4. Replacing these SHAP-selected features also produced larger mean normalised prediction changes than replacing random subsets of the same size.

Table 4: Feynman output completeness at 90 percent absolute attribution coverage on 50 fresh final observations

Model	**Mean features**	**Mean normalised preservation error SHAP / random**	**Mean normalised deletion change SHAP / random**	**Median sustained reconstruction features**
Random Forest	4.64	0.3027 / 0.4155	0.9049 / 0.8479	6
XGBoost	4.42	0.3665 / 0.5073	0.9028 / 0.8589	6
MLP	4.54	0.3104 / 0.4908	0.9530 / 0.8800	6

Preservation and deletion values are dimensionless: each observation's curve is divided by its own mean absolute prediction change when all features are replaced, before averaging across observations. Lower preservation error and higher deletion change indicate stronger performance under the stated replacement procedure. Figure 1 shows the mean curves over all subset sizes. The SHAP ordering generally reduced preservation error earlier and increased deletion change earlier than the random ordering.

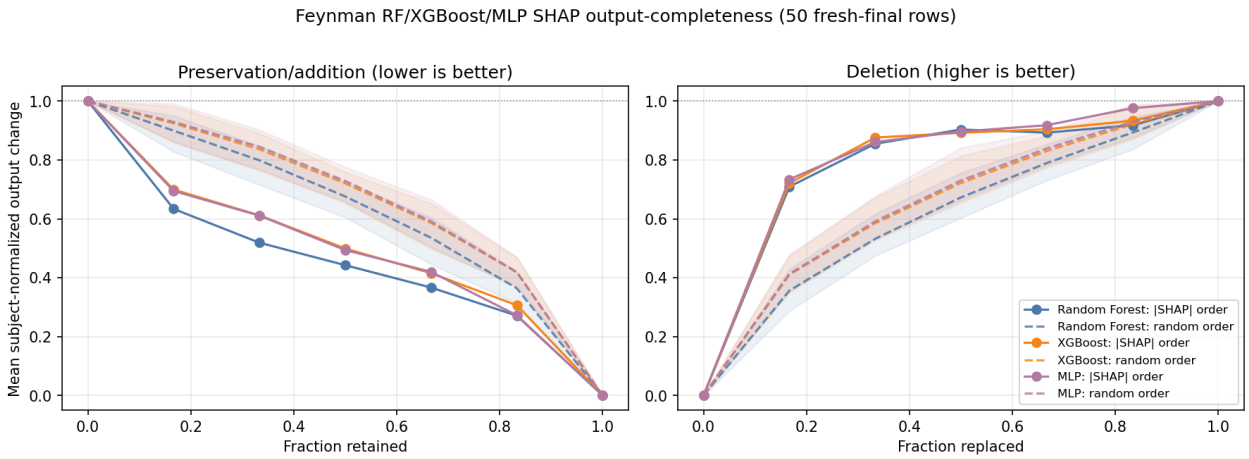


Figure 1: Feynman preservation and deletion curves for 50 fresh final observations. Solid lines use decreasing absolute SHAP order; dashed lines use matched random orders. Shading shows empirical 2.5th to 97.5th percentiles across random order repeats. These ranges describe the random reference, rather than population confidence intervals.

At the 90%-attribution prefix, sustained reconstruction was satisfied in only 3/50 Random

Forest cases, 1/50 XGBoost cases and 1/50 MLP cases. The median sustained reconstruction count was all six features for every model, using tolerances of approximately 0.0191, 0.0198 and 0.0198 dimensionless units for Random Forest, XGBoost and MLP, respectively. Individual curves were not always monotonic: preservation reversals occurred in 36, 38 and 38 observations, and deletion reversals in 48, 47 and 48. Sustained reconstruction required the tolerance to be satisfied at the selected prefix and every larger prefix. This replacement test is separate from full-vector SHAP additivity, which checks that the baseline plus all attributions reproduces the original fitted prediction and passed its numerical tolerance checks. No observation had undefined normalisation.

For PySR, native-rule completeness (reasoning-completeness in Section 3.5.3) concerned disclosure of the fitted equation, its dimensionless inputs and direct output mapping. These specify how its prediction is calculated, separately from reconstruction under feature replacement.

4.4 Continuity

SHAP explanations remained similar among perturbed observations whose fitted predictions met the response-stability condition. Table 5 reports results for 50 historical test observations and 20 perturbation repeats per noise level. The condition allowed a prediction change of at most approximately 0.01985 dimensionless units. Each model and noise level therefore had 1,000 candidate pairs, of which different numbers were retained.

Table 5: Feynman SHAP continuity among response stable pairs with attribution distance in dimensionless target units per feature

Model	**Noise percent SD**	**Retained / 1000**	**Mean cosine**	**Mean attribution distance**
Random Forest	2.5	865	0.999391	0.002474
Random Forest	5.0	660	0.998029	0.004298
XGBoost	2.5	724	0.999145	0.002957
XGBoost	5.0	517	0.997532	0.005390
MLP	2.5	917	0.999361	0.003186
MLP	5.0	626	0.998018	0.005808

Mean cosine similarities ranged from 0.997532 to 0.999391, indicating little change in the direction of the signed attribution vectors among retained pairs. Mean absolute attribution distance, measured in target units per feature, increased at the higher noise level for every model. Fewer pairs also met the response-stability condition at 5% noise. High cosine similarity consequently describes the retained cases and does not establish stability for all perturbed inputs. Independent SHAP recalculations on unchanged inputs returned identical vectors for all 50 checked observations per model.

Table 6 reports the separate prediction-noise experiment for all four predictors over 100 repeats per level on the 2,000 historical test rows. Noise was 2.5% or 5% of each training-feature standard deviation. Mean RMSE averages the repeat-specific errors against unchanged clean targets; mean absolute prediction change averages absolute differences from each row's clean

fitted prediction across rows and repeats. These historical diagnostics are distinct from the fresh-final performance in Table 2 and the 50-row, 20-repeat SHAP continuity test. Prediction changes increased at the higher noise level for every model; these results do not establish a general robustness ranking.

Table 6: Feynman prediction noise results on 2,000 historical test rows with dimensionless errors and changes

Model	**Noise percent training SD**	**Mean RMSE**	**Mean absolute prediction change**
Random Forest	2.5	0.082945	0.009302
Random Forest	5.0	0.084478	0.017378
XGBoost	2.5	0.034962	0.013069
XGBoost	5.0	0.039548	0.021643
MLP	2.5	0.012574	0.008425
MLP	5.0	0.022967	0.016850
PySR	2.5	0.011117	0.008442
PySR	5.0	0.022237	0.016885

The native PySR rule remained structurally fixed, while its predictions changed with the inputs. All 200,000 perturbed evaluations per level remained finite. Minimum equation denominators were 0.437340 and 0.426447, with no near-zero denominator detected. These checks support numerical regularity over the tested inputs; unchanged equation structure does not provide a SHAP-vector continuity score.

4.5 Compactness

At 90% absolute attribution coverage, the mean local explanation contained 4.68 features for Random Forest and 4.52 for both XGBoost and MLP. All three had a median of five features and an interquartile range from four to five over the 2,000 historical test observations. Table 7 also shows the sensitivity to the coverage threshold. Mean counts increased from approximately four features at 80% coverage to just over five at 95% coverage.

Table 7: Feynman local SHAP feature counts over 2,000 historical test observations

Model	**Mean at 80%**	**Mean at 90%**	**Mean at 95%**	**Median at 90%**	**Range at 90%**
Random Forest	3.81	4.68	5.26	5	3 to 6
XGBoost	3.68	4.52	5.13	5	2 to 6
MLP	3.67	4.52	5.14	5	2 to 6

No observation had zero total attribution. These counts describe the size of an attribution summary under the chosen coverage convention. They differ slightly from the counts in Table 4 because compactness used the full historical test sample, while completeness used 50 fresh final observations.

The literal saved PySR expression contained six distinct variables, ten operator occurrences and one numeric constant leaf, giving 17 expression-tree nodes. This matched its recorded PySR complexity. The count includes the original expression's redundant squaring of the constant 1, which disappears in the simplified equation displayed above. The global equation size and local attribution counts use different units and do not provide a common compactness ranking.

5 Body Fat Results

The body-fat results concern 51 test subjects from the observational dataset. Unlike the Feynman case, there is no known generating equation against which recovery can be assessed. The explanation results therefore concern the fitted models and their prediction rules. Primary SHAP evaluations use the 13 raw measurements, with both derived ratios recomputed inside the prediction wrapper. The chapter also reports the supplementary feedback from two survey respondents.

5.1 Predictive performance

MLP had the lowest test error, followed by PySR, XGBoost and Random Forest when ordered by RMSE. Table 8 shows an MLP RMSE of 3.737 body-fat percentage points and a PySR RMSE of 3.883, compared with approximately 4.12 to 4.13 for the tree models. MLP also had the lowest MAE and highest R^2. Random Forest had slightly lower MAE than XGBoost despite its slightly higher RMSE. These are descriptive results on one 51-subject test split; they do not establish a general performance ranking.

Table 8: Body fat predictive performance on 51 test subjects

Model	R^2	**RMSE (percentage points)**	**MAE (percentage points)**
Random Forest	0.633	4.130	3.440
XGBoost	0.635	4.121	3.517
MLP	0.700	3.737	3.110
PySR	0.676	3.883	3.256

5.2 Correctness

5.2.1 Native prediction rule and SHAP checks

The validation-selected PySR equation was:

$$\widehat{\text{BodyFat}} = \frac{\text{Abdomen}^2 - 272.58493\,\text{Wrist}}{\text{Weight} + \text{Neck} - \text{Biceps}}.$$

Weight is measured in pounds and the displayed circumferences in centimetres. The output is body-fat percentage. The equation uses five of the raw measurements and neither derived ratio. Evaluation with the printed constant reproduced saved predictions to a maximum absolute difference of 3.553×10^{-15} percentage points, within the specified 10^{-12} tolerance. This confirms that the disclosed rule reproduces the fitted symbolic predictor. Without a known generating relationship, it does not establish recovery of the process underlying body composition.

The primary SHAP results used exact enumeration of all 8,192 coalitions against the fixed 50-row training background. Independent calculations returned identical values for all 51 subjects for the tree models and for the 12 checked MLP subjects. The earlier one-cycle permutation estimates were therefore superseded for the computational evaluations reported here. These checks remove permutation-sampling variation for this background, while leaving the interpretation dependent on the chosen background and raw-feature representation.

5.2.2 Local and global fitted model faithfulness

Local SHAP rankings generally agreed with the prediction impacts obtained by replacing each raw measurement using all 201 training donors. Median Spearman correlation was 0.7308 for Random Forest, 0.8462 for XGBoost and 0.8791 for MLP (Table 9). Correlations were defined for all 51 subjects per model. MLP had the highest median and the narrowest interquartile range, although agreement varied within every model.

Table 9: Body fat local ranking agreement on 51 test subjects

Model	**Median ρ**	**Q1**	**Q3**	**Median top three overlap**
Random Forest	0.7308	0.6264	0.8269	0.6667
XGBoost	0.8462	0.7143	0.8984	0.6667
MLP	0.8791	0.7967	0.9423	0.6667

All models had a median overlap of two of the three highest-ranked features. Complete top-three agreement occurred for 15 Random Forest subjects, 19 XGBoost subjects and 25 MLP subjects. The lowest local correlations were 0.1154, 0.4945 and 0.3681, respectively. Aggregate agreement consequently did not imply close agreement for every subject.

In the global summaries, correlations between mean absolute SHAP and mean deletion impact were 0.9615 for Random Forest, 0.9286 for XGBoost and 0.9890 for MLP. Abdomen was the highest-ranked measurement under both summaries for every model. Its mean absolute SHAP value was 4.552, 4.865 and 5.383 percentage points, respectively. The remaining SHAP rankings differed: Random Forest placed Chest and Height next, XGBoost placed Wrist and Age next, and MLP placed Wrist and Chest next. These are properties of the fitted predictors under the stated explanation and replacement procedures. Recomputing the ratios preserved the deterministic feature engineering, but donor replacement could still produce anatomically unrealistic combinations.

5.2.3 Supplementary native feature importance

The separate comparison over 15 prepared inputs gave native-importance versus SHAP rank correlations of 0.821 for Random Forest and 0.629 for XGBoost. Top-three overlap was 1.000 and 0.667, respectively. Both measures placed Abdomen first in both models. One substantial disagreement concerned XGBoost Weight, ranked second by native gain but thirteenth by mean absolute SHAP. This comparison used direct TreeSHAP on the 15 model inputs, so it remains separate from the primary 13-measurement exact-SHAP results.

5.3 Completeness

SHAP-selected subsets preserved body-fat predictions substantially better on average than random subsets of the same size. At 90% absolute attribution coverage, mean normalised preservation errors were 0.1003 for Random Forest, 0.1247 for XGBoost and 0.1629 for MLP, compared with random-reference errors of 0.5629, 0.6403 and 0.6619 (Table 10). Deleting the selected features also produced larger changes than the matched random replacements.

Table 10: Body fat output completeness at 90 percent absolute attribution coverage on all 51 test subjects

Model	**Mean features**	**Mean normalised preservation error SHAP / random**	**Mean normalised deletion change SHAP / random**	**Median sustained reconstruction features**
Random Forest	6.35	0.1003 / 0.5629	0.9679 / 0.5514	11
XGBoost	5.94	0.1247 / 0.6403	1.0029 / 0.5642	10
MLP	6.43	0.1629 / 0.6619	0.9744 / 0.6588	11

The complete curves in Figure 2 show a large mean deletion change after replacing the first ranked measurement, with smaller additional changes as more features were replaced. Preservation error also fell most sharply at the first few retained measurements. As in Section 4.3, each subject's curve is normalised by its own mean absolute all-replaced prediction change before averaging. These values are dimensionless and are not bounded by one. XGBoost's mean deletion value of 1.0029 therefore indicates a slightly larger change than its all-replaced reference at the selected prefixes.

At the 90%-attribution prefix, sustained reconstruction was satisfied in only 1/51 Random Forest cases, 2/51 XGBoost cases and 2/51 MLP cases. For MLP, 4/51 subjects met tolerance at the selected prefix alone, but only 2/51 remained within tolerance at every larger prefix. Median sustained reconstruction counts were 11 features for Random Forest, 10 for XGBoost and 11 for MLP, compared with mean attribution-coverage counts of approximately six. Reconstruction tolerances were 0.3740, 0.3702 and 0.3561 percentage points, respectively. Mean raw preservation errors at the 90% subsets were 0.7630, 0.9511 and 1.1288 percentage points. Thus, the reduced summaries retained influential information while leaving prediction differences larger than these tolerances on average.

Preservation curves reversed direction for 48 Random Forest subjects and all 51 subjects for XGBoost and MLP. Deletion reversals occurred for 50, 51 and 51 subjects. No subject



Figure 2: Body fat preservation and deletion curves for all 51 test subjects. Solid lines use decreasing exact absolute SHAP order; dashed lines use matched random orders. Shading shows empirical 2.5th to 97.5th percentiles across random order repeats. Both ratios are recomputed after replacement.

had undefined normalisation. These results support retaining the sustained reconstruction definition, rather than treating the first tolerance crossing as sufficient.

For PySR, the five-variable equation and specified input preparation supply native-rule completeness by disclosing how the fitted output is calculated. The equation's adequacy as a description of body composition remains unresolved.

5.4 Continuity

The exact-SHAP continuity test used 12 selected test subjects and ten perturbation repeats per noise level, giving 120 candidate pairs per model and level. Pairs were retained when the fitted prediction changed by no more than approximately 0.433743 percentage points. Table 11 shows high mean cosine similarity among retained pairs, ranging from 0.997993 to 0.999134. Mean absolute attribution distance increased at the higher noise level for each model.

Table 11: Body fat exact SHAP continuity among response stable pairs with attribution distance in percentage points per feature

Model	**Noise percent SD**	**Retained / 120**	**Mean cosine**	**Mean attribution distance**
Random Forest	2.5	104	0.999134	0.017219
Random Forest	5.0	90	0.998780	0.025331
XGBoost	2.5	83	0.999087	0.015143
XGBoost	5.0	74	0.997993	0.024982
MLP	2.5	110	0.999049	0.023117
MLP	5.0	78	0.998230	0.037296

Attribution distance is measured in body-fat percentage points per feature. The retained counts declined from 104 to 90 for Random Forest, 83 to 74 for XGBoost and 110 to 78 for MLP when noise increased from 2.5% to 5% of training-feature standard deviation. Each model could retain different pairs, so the means concern different conditional subsets. Sensitivity checks using 2.5% and 10% of training-response standard deviation as the response threshold retained the high-cosine pattern while changing the retained counts. These results support local continuity in these tested cases, with its scope limited by the 12-subject, model-informed subset and fixed background.

Table 12 reports the separate 100-repeat prediction-noise experiment on all 51 test subjects, with noise at 2.5% or 5% of each raw training-feature standard deviation and both ratios recomputed. Mean RMSE uses unchanged clean targets, and mean absolute prediction change compares perturbed and clean fitted predictions, averaging as in Section 4.4. This experiment is distinct from clean performance in Table 8 and the 12-subject, ten-repeat SHAP continuity test. Mean absolute prediction change increased for every model at the higher noise level.

Table 12: Body fat prediction noise results on all 51 test subjects with errors and changes in percentage points

Model	**Noise percent training SD**	**Mean RMSE**	**Mean absolute prediction change**
Random Forest	2.5	4.132466	0.226715
Random Forest	5.0	4.163985	0.445370
XGBoost	2.5	4.062050	0.289188
XGBoost	5.0	4.109880	0.515257
MLP	2.5	3.748462	0.181421
MLP	5.0	3.773600	0.362743
PySR	2.5	3.897623	0.222748
PySR	5.0	3.931814	0.445580

The native PySR equation was unchanged across the same 12-subject perturbations, but its mean absolute prediction change increased from 0.2297 to 0.4596 percentage points. It retained 105 of 120 response-stable pairs at 2.5% noise and 65 at 5% noise. Across the wider domain check of 5,100 perturbed rows per level, all raw measurements remained positive and all prepared inputs and PySR predictions remained finite. No near-zero equation denominator was detected; minimum absolute denominators were 147.8902 and 145.9304. However, 200 and 209 rows contained a training-range violation. These finite-sample checks describe the evaluated domain rather than guaranteeing stability elsewhere.

5.5 Compactness

The 90% absolute-attribution summaries used about six of the 13 raw measurements on average. XGBoost had the lowest mean count at 5.94, followed by Random Forest at 6.35 and MLP at 6.43. Median counts were six, six and seven, respectively. Table 13 shows that

increasing the coverage threshold to 95% raised mean counts to approximately eight, while 80% coverage required approximately four. No subject had zero total attribution for any model.

Table 13: Body fat local SHAP feature counts over 51 test subjects

Model	**Mean at 80%**	**Mean at 90%**	**Mean at 95%**	**Median at 90%**	**Range at 90%**
Random Forest	3.96	6.35	8.33	6	3 to 10
XGBoost	4.12	5.94	7.84	6	3 to 8
MLP	4.45	6.43	8.20	7	3 to 9

The literal native PySR expression used five distinct variables, six operator occurrences and one numeric constant leaf, totalling 12 expression-tree nodes. This matched its recorded complexity. These counts describe the global rule and remain separate from local attribution-coverage counts.

5.6 Exploratory survey

Two respondents completed the body-fat survey, both reporting expertise in statistics, data science or machine learning. All 36 ratings and six preference answers were recorded. Table 14 presents individual ratings for the Random Forest and XGBoost beeswarms and PySR equation. MLP was not included.

Table 14: Individual survey ratings on a five point agreement scale

Rating item	**Random Forest R1 / R2**	**XGBoost R1 / R2**	**PySR R1 / R2**
Understanding how to read and evaluate	5 / 5	5 / 5	5 / 5
Plausibility of emphasised measurements	2 / 4	4 / 4	4 / 4
Plausibility of directions of influence	3 / 4	4 / 4	5 / 5
Understanding higher or lower measurement effects	2 / 3	3 / 3	5 / 5
Appropriate amount of information	4 / 5	4 / 5	5 / 5
Usefulness for review or communication	4 / 4	4 / 4	4 / 5

Ratings ranged from 1, Strongly Disagree, to 5, Strongly Agree; item labels in Table 14 are shortened. Both respondents selected PySR as easiest to interpret, most consistent with their body-composition knowledge, and preferred for reviewing a prediction. Thus, PySR received both responses on each of the three preference questions. All three explanations received the highest rating for understanding how to read and evaluate them. The clearest rating difference concerned understanding how higher or lower measurement values would affect a prediction: PySR received 5 from both respondents, compared with 2 and 3 for Random Forest and 3 and 3 for XGBoost.

The stated reasons for preference were discovering more factors relevant to body fat and clearer directions of feature effects. Neither respondent supplied an optional comment about implausible, misleading or unclear content; this does not establish that no problems were perceived.

These two responses describe perceptions of the presented model-explanation combinations. The archived September form export postdates the August responses, so it cannot establish precisely which version respondents saw. Presentation order was fixed, and the beeswarms used historical permutation-SHAP values with supplementary univariable p-value labels. The feedback therefore concerns different stimuli from the final exact-SHAP results and establishes neither demonstrated understanding nor representative user preferences.

6 Discussion

6.1 Overview of the findings

Across the two datasets, SHAP and PySR provided different kinds of information about their fitted models. SHAP supplied signed feature contributions to individual Random Forest, XGBoost and MLP predictions, while PySR supplied the mathematical rule used to calculate its predictions. The evaluations of correctness, completeness, continuity and compactness showed the strengths and limitations of these explanation forms. SHAP generally identified influential inputs, but shortened attribution summaries usually omitted information needed for close prediction recovery. PySR disclosed an executable rule, with the Feynman case also allowing recovery of the generating relationship to be verified.

Predictive performance provided context for this comparison. PySR recovered the known Feynman equation and reproduced its outputs to numerical precision on the fresh final sample (Table 2). On body fat, MLP had the lowest observed RMSE, followed by PySR, at 3.737 and 3.883 percentage points on 51 test subjects (Table 8). The explicit equation therefore performed reasonably close to the lowest observed error in this case. The known Feynman relationship permits a direct check of what was recovered. The observational body-fat data provide no equivalent reference for the retained measurements, so their explanation evidence concerns the fitted predictors and their behaviour.

6.2 Correctness

The local SHAP rankings generally agreed with the single-feature replacement impacts. Median Spearman correlations were approximately 0.66–0.77 for Feynman and 0.73–0.88 for body fat (Tables 3 and 9). Larger absolute contributions generally identified inputs whose replacement caused larger prediction changes. The median top-three overlap was two features for every model in both datasets. This supports the usefulness of the rankings under the replacement procedure, although agreement varied between observations. Some Feynman tree-model cases had negative correlations, showing that the aggregate pattern did not apply to every explanation.

The two rankings also answer somewhat different questions. SHAP averages marginal contributions across feature coalitions, whereas single-feature replacement changes one input while retaining the other observed values [4]. Their rankings need not match exactly.

Interactions and dependent inputs are possible reasons for differences, but their separate effects were not established by this study. A rank correlation measures agreement between orderings; it is not the proportion of an explanation that is correct. Agreement in absolute ranks also does not test the signs of contributions. The body-fat global summaries provided a concrete example of agreement: abdomen ranked first under both mean absolute SHAP and mean replacement impact for every predictor. This describes the fitted models rather than establishing a biological effect.

For PySR, correctness required distinguishing execution of the printed rule from recovery of a generating relationship. Evaluating the disclosed expressions reproduced the saved predictions in both datasets. This verified that the equations and their specified inputs represented the fitted symbolic predictors. Feynman supplied an additional check: subtracting the known expression from the selected equation and simplifying gave zero. Algebraic equivalence established recovery on the non-singular domain, beyond the evidence supplied by low prediction error alone.

The body-fat audit established reproduction of the fitted rule, but could not establish an underlying biological equation. This distinction follows the separation between faithfulness to a predictor and agreement with the phenomenon being modelled discussed in the Background [9]. The evidence differed because only Feynman provided a known reference relationship.

6.3 Completeness

SHAP-selected subsets preserved predictions better on average than matched random subsets in both datasets. Replacing the selected features also caused larger mean changes than replacing equally sized random subsets (Tables 4 and 10). The ordering therefore contained useful information about the fitted predictions. For body-fat MLP, mean normalised preservation error at the 90%-attribution subset was approximately 0.163, compared with 0.662 for the random reference. These are replacement errors normalised by each subject's mean absolute all-replaced prediction change, rather than percentages of predictive accuracy. Figures 1 and 2 show how preservation and deletion changed as more features were included.

However, retaining most attribution magnitude usually left out input information needed for close prediction recovery. For body fat, median feature counts at 90% absolute attribution coverage were six, six and seven for Random Forest, XGBoost and MLP (Table 13). Median sustained reconstruction counts were eleven, ten and eleven. Reconstruction compared each observation's mean absolute prediction change across donor replacements with a tolerance of 5% of each model's training-prediction standard deviation. Sustained reconstruction required this mean change to remain within tolerance at the selected SHAP-ranked prefix and every larger prefix. The reduced summaries captured the largest contributions, but usually did not retain enough observed inputs to keep this mean prediction change within tolerance when the remaining measurements were replaced. Influential information and sufficient information for close recovery were consequently different outcomes.

The same distinction occurred in Feynman. On the 50 fresh final observations, coverage subsets contained approximately 4.4–4.6 features on average, while median sustained reconstruction required all six inputs (Table 4). The gap between coverage and recovery therefore also appeared in the controlled known-equation setting. Individual preservation curves sometimes reversed direction as further features were retained. Applying this crite-

rion to SHAP-ranked prefixes avoided treating a temporary crossing as sustained recovery; the counts did not involve a search over every possible feature subset. The predictors were held fixed throughout these tests. The reconstruction counts describe the inputs needed to preserve their predictions under replacement; they do not establish how many inputs a newly trained predictor would need.

These results are compatible with SHAP's additive representation. The baseline plus the full signed attribution vector reproduced the original fitted output within the numerical checks, consistent with local accuracy [4]. The subset experiment instead evaluated new predictions after omitted inputs had been replaced. Covering most absolute attribution magnitude at the original observation does not specify how the predictor will respond to those new combinations. The reconstruction results therefore concern the information retained under replacement, rather than a failure of full-vector additivity.

PySR supplied a different form of completeness. Its equation, fitted constants and required input definitions disclosed the prediction calculation. This corresponds to reasoning-completeness for the fitted symbolic model, distinct from the SHAP subset experiment [9]. The Feynman rule required the specified dimensionless inputs. The body-fat rule used five raw measurements and neither engineered ratio; its stated units and preparation remained necessary for reproduction. An executable rule exposes how its model calculates an output, while the subset tests assess whether selected inputs preserve another model's prediction. These forms of evidence cannot be treated as equivalent completeness scores.

6.4 Continuity

SHAP attribution directions changed little among retained nearby pairs in both datasets. Mean cosine similarities across models and noise levels were approximately 0.9975–0.9994 (Tables 5 and 11). The corresponding attribution distances were non-zero and increased at the higher noise level for every model. The contributions therefore remained similar in direction while their values changed. Considering both measures gives a fuller account of continuity than cosine similarity alone.

This finding was conditional on similar fitted responses. Pairs were retained only when the prediction changed by no more than 5% of the training-target standard deviation. For example, body-fat XGBoost retained 74 of 120 candidate pairs at the higher noise level. Its high cosine similarity describes those retained pairs. Different predictors could retain different pairs, so the small differences between their cosine means do not provide a useful model ranking. The explanation-continuity tests used smaller subsets of the same observations and perturbation repeats from the larger prediction-noise experiments reported in Tables 6 and 12.

PySR output sensitivity required separate interpretation. A fixed expression can still produce changing predictions when its inputs change. Across the tested perturbations, outputs remained finite and no near-zero equation denominator was detected in either dataset. These checks support numerical regularity over the evaluated inputs, without supplying a SHAP-vector continuity score. For Feynman, perturbing inputs also changes the known equation's output. Error against unchanged clean targets in the prediction-noise experiment therefore does not indicate failure to recover that equation.

6.5 Compactness and reported understanding

The size of a SHAP summary depended on the attribution coverage requested. At 90% coverage, median Feynman summaries contained five of six features on the historical test sample (Table 7). Body-fat medians were six, six and seven of thirteen for Random Forest, XGBoost and MLP (Table 13). Raising the coverage threshold increased the feature counts in both datasets. Compactness therefore depended on a reporting choice about how much attribution to retain. The completeness results show why a shorter display should not automatically be taken as sufficient for close prediction recovery.

PySR structure described the size of a global calculation. The literal Feynman expression contained six variables and 17 expression-tree nodes, while the body-fat expression contained five variables and 12 nodes. These counts concern the saved expressions, including structure removed by simplifying the Feynman display. A global equation and a local attribution summary present different information, so their node and feature counts cannot be ranked directly as measures of understanding. As discussed in SRBench++, expression size is an incomplete proxy for interpretability [12].

The survey provided limited feedback about how the body-fat explanations were received. Both respondents preferred PySR, and both rated their understanding of higher or lower measurement effects at 5/5 for the equation (Table 14). These responses describe reported understanding from two people. They give some support to the accessibility of the equation shown, but do not establish demonstrated understanding or representative user preferences. Answering questions about predictions would provide different evidence from rating an explanation.

6.6 Practical implications

SHAP is useful when the aim is to examine individual predictions from an existing fitted model. Its signed contributions show which features contributed towards predictions above or below the explainer's baseline. The replacement results support using the absolute rankings to identify influential inputs in these cases. A shortened display can communicate the largest contributions, but its scope should be clear. If a user needs enough retained input information for close prediction recovery, that requirement needs its own check. Choosing a high attribution-coverage threshold alone did not meet it in most of the tested observations.

PySR offers an explicit calculation that can be inspected and reproduced. Feynman demonstrated the value of this form when a recovered expression could be verified against a known relationship. Body fat illustrated a five-measurement fitted rule with predictive performance reasonably close to the lowest observed test error. The equation made its operations and constant available for inspection, although this still required a reader to follow the calculation and its units. Its transparency concerns the model it represents; predictive adequacy and ease of understanding remain separate considerations. SHAP can describe an already selected predictor, whereas obtaining a native PySR equation involves selecting a symbolic predictor as well.

The comparison also shows the need for better ways to assess different explanation forms. The four properties organised the evaluation, but some measurements answered different questions for SHAP and PySR. A shared task, such as identifying which specified input

change produces the largest prediction effect, could supply a common outcome. This would allow the explanations to be assessed by what they help a reader do, without placing equation nodes and attribution counts on one numerical scale. The present results support different uses for the two approaches, rather than a universal winner or a combined explanation-quality score.

6.7 Limitations

The comparison changed the predictor as well as the explanation form. SHAP described Random Forest, XGBoost and MLP, while the native equation described PySR. Differences therefore belonged to these model–explanation combinations and could not be attributed solely to presentation. Feynman also used path-dependent TreeSHAP for the trees and empirical-background exact SHAP for MLP. Training samples further qualified the body-fat performance comparison: the selected PySR expression used 150 fitting subjects and a separate validation set, while the other final predictors used all 201 training subjects.

The empirical scope was limited to one Feynman problem and one observational body-fat dataset. Feynman used prepared dimensionless inputs and an operator set able to express the known relationship. Its recovery result concerns that setup. The body-fat dataset contained 252 men, with predictive performance assessed on a single 51-subject test split. No repeated-split comparison established whether the observed error difference between MLP and PySR would persist under another allocation of training and test subjects. Its explanation-continuity test involved only 12 selected subjects. Repeated perturbations supplied additional checks on those subjects, rather than additional independent subjects. These samples limit conclusions about other problems, populations and data splits.

The explanation and replacement procedures also affected the interpretation. Primary body-fat SHAP calculations used one fixed empirical background. Exact enumeration removed coalition-sampling approximation for that background, but did not establish independence from background selection. Donor replacements could break relationships between observed and replaced measurements. Recalculating the ratios maintained the feature engineering without ensuring realistic hybrid measurements. Feynman replacements could likewise break relationships between ratios sharing a denominator. The ranking and reconstruction findings consequently describe behaviour under the specified replacements.

Human understanding received little direct assessment. Computational checks of the four properties did not establish how successfully people could use the explanations. The survey had two respondents, a fixed presentation order and historical SHAP stimuli rather than the final exact explanations. It also omitted MLP, and the archived form did not establish the precise version seen by respondents. These constraints give the survey limited weight in judging explanation quality. Reported preference usefully complements the computational findings, but cannot determine a general choice between SHAP and PySR.

6.8 Future work

A larger task-based user study would directly assess what people can learn from each explanation form. Participants could identify whether a specified measurement change raises a prediction, or which of several changes has the largest effect. Answer accuracy and time

could be measured alongside preference, with presentation order randomised. One useful design would compare the native PySR equation with SHAP applied to the same PySR predictor. Holding the model and cases fixed would help separate the effect of explanation form from differences between predictors. Correct answers could be calculated directly from that predictor, providing an objective reference for the participants' responses. This would connect evaluation to a defined use of an explanation, while retaining preference as a separate outcome.

Further computational work could repeat the analysis with several background samples and replacements that better preserve relationships between measurements. This would examine how much the ranking and reconstruction findings depend on the present choices. Continuity tests could include more observations and consider explanation change alongside prediction change across all perturbations, as well as within the response-stable subset.

Additional known-equation problems, including noisy observations and less prepared inputs, would extend the recovery evidence beyond the single Feynman relationship. More observational datasets and repeated data splits would also help assess whether the explanation patterns and small body-fat performance differences persist.

The findings address the study aim by showing what each explanation form made available for inspection. SHAP supplied informative local contributions, while PySR exposed a fitted prediction rule and enabled verified recovery where a reference equation existed. The distinction between attribution coverage and close prediction recovery remains central to interpreting the shortened SHAP summaries.

7 Conclusion

This study compared SHAP explanations of Random Forest, XGBoost and MLP models with equations produced by PySR. Correctness, completeness, continuity and compactness were examined on a dimensionless Feynman benchmark and an observational body-fat dataset, with predictive performance providing context. The comparison showed what each approach revealed about its fitted model and where the available evidence limited its interpretation.

PySR recovered the known Feynman generating equation. Algebraic equivalence established recovery wherever the expressions were defined, while prediction error reached numerical precision on the fresh final sample. The body-fat case provided a different kind of evidence. PySR produced an explicit equation using five measurements, with an RMSE of 3.883 percentage points compared with 3.737 for MLP, the lowest observed error on the 51 test subjects. The equation therefore achieved predictive performance reasonably close to the lowest error in this case. However, the observational data provided no known generating relationship, so the equation remained a description of the fitted predictor.

The SHAP evaluations supported the usefulness of its feature rankings under the specified replacement procedure. Features with larger absolute contributions generally caused larger prediction changes when replaced, and SHAP-selected subsets preserved predictions better on average than matched random subsets. Under small input perturbations, attribution vectors also remained highly similar in direction among pairs whose predictions changed little. The attribution values still changed, and the continuity result applied to the retained pairs. These findings support the local information supplied by SHAP within the evaluated settings.

The clearest limitation of shortened SHAP summaries concerned prediction recovery. For body fat, median feature counts covering 90% of absolute attribution were six, six and seven for Random Forest, XGBoost and MLP. The corresponding medians for sustained close prediction recovery were eleven, ten and eleven. Recovery required each subject's mean absolute prediction change across donor replacements to remain within the chosen tolerance at the selected subset and every larger subset in SHAP order. A similar gap occurred in Feynman, where median sustained recovery required all six inputs. Thus, retaining most attribution magnitude usually omitted information needed to closely preserve the fitted prediction under replacement. This result concerned new predictions after omitted inputs were replaced. The baseline and full signed SHAP vector still accounted for the original output within numerical tolerance.

The PySR equations, together with their constants, input definitions and preparation, disclosed the complete fitted prediction calculation. This made the rule available for direct inspection and reproduction. Its compactness described the structure of a global equation, while SHAP compactness described the number of features retained in a local summary. These sizes did not provide a common measure of understanding. Likewise, a fixed equation and similar SHAP vectors supplied different evidence about behaviour under changing inputs. The four properties therefore helped organise the comparison, but their measurements could not be combined into one explanation-quality score.

The study contributed an empirical comparison of local attributions and explicit prediction rules, and a practical application of four explanation properties to these different forms. It also identified where those evaluations answered different questions. The distinction between attribution coverage and prediction recovery was a central result, showing why the information required from an explanation should be stated explicitly.

These conclusions remain limited to the two datasets and evaluation procedures used. Body-fat performance was assessed on one test split, and the comparison changed the predictor as well as the explanation form. Differences therefore could not be attributed to presentation alone. Background selection and potentially unrealistic donor replacements also limited the interpretation of the SHAP results. Both survey respondents preferred the presented PySR equation, but two responses about reported understanding and preference did not establish how successfully people could use the explanations more broadly.

Further work should directly test what users can do with each explanation while holding the predictor and cases fixed. More datasets, repeated data splits and checks of background and replacement choices would also help establish how widely the observed patterns apply.

For the settings examined, SHAP supported inspection of individual predictions from an existing model, while PySR exposed the global rule of a symbolic predictor. Choosing between them depends on whether the task requires local feature contributions or an explicit calculation, alongside acceptable predictive performance. Evaluation should establish how well an explanation supplies the required information and how successfully people can use it.

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